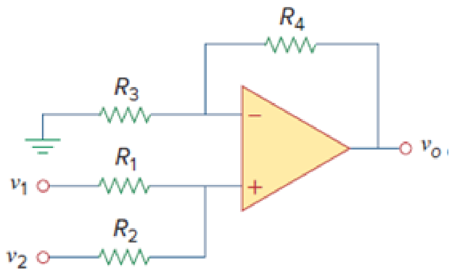
Chapter 5, Problem 44P

Problem

Show that the output voltage *vo* of the circuit in Fig. 5.78 is

Figure. 5.78:

$$v_o = \frac{(R_3 + R_4)}{R_3(R_1 + R_2)}(R_2 v_1 + R_1 v_2)$$



Step-by-step solution

Step 1 of 4

Refer to Figure 5.78 in the textbook.

Draw the circuit diagram with node notations.

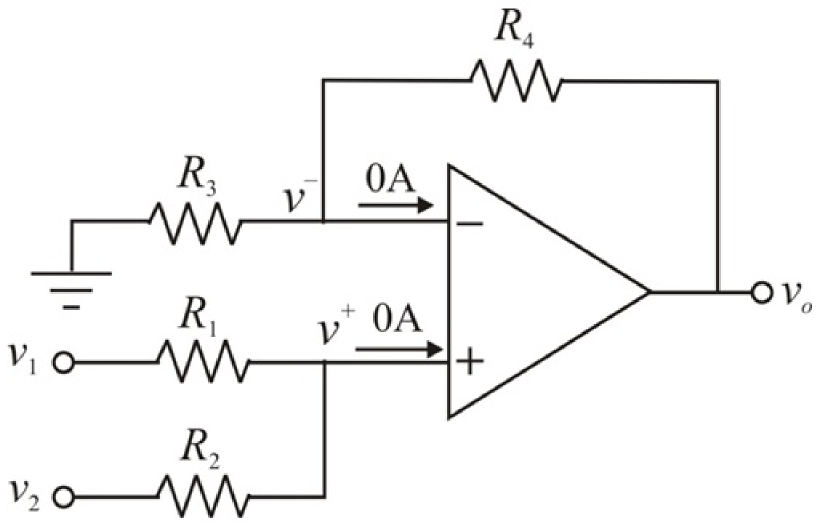


Figure 1

Step 2 of 4

In an ideal op-amp, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal.

$$v^- = v^+$$

Apply Kirchhoff's current law at node v^-.

$$\frac{v^-}{R_3} + \frac{v^- - v_o}{R_4} = 0$$

$$v^-\left[\frac{1}{R_3} + \frac{1}{R_4}\right] - \frac{v_o}{R_4} = 0$$

$$v^-\left[\frac{R_4 + R_3}{R_3 R_4}\right] = \frac{v_o}{R_4}$$

$$v^-\left[\frac{R_3 + R_4}{R_3}\right] = v_o$$

$$v^- = v_o\left[\frac{R_3}{R_3 + R_4}\right]$$

Step 3 of 4

Apply Kirchhoff's current law at node v^+.

$$\frac{v^+ - v_1}{R_1} + \frac{v^+ - v_2}{R_2} = 0$$

$$\frac{R_2\left(v^+ - v_1\right) + R_1\left(v^+ - v_2\right)}{R_1 R_2} = 0$$

$$R_2\left(v^+ - v_1\right) + R_1\left(v^+ - v_2\right) = 0$$

$$R_2 v^+ - R_2 v_1 + R_1 v^+ - R_1 v_2 = 0$$

$$\left(R_1 + R_2\right)v^+ - R_2 v_1 - R_1 v_2 = 0$$

$$\left(R_1 + R_2\right)v^- - R_2 v_1 - R_1 v_2 = 0 \qquad \left(\text{Since, } v^- = v^+\right)$$

Step 4 of 4

Substitute $v_o\left[\frac{R_3}{R_3 + R_4}\right]$ for v^-.

$$v_o\left[\frac{R_3\left(R_1 + R_2\right)}{R_3 + R_4}\right] - R_2 v_1 - R_1 v_2 = 0$$

$$v_o\left(\frac{R_3\left(R_1 + R_2\right)}{R_3 + R_4}\right) = R_2 v_1 + R_1 v_2$$

$$v_o = \frac{\left(R_3 + R_4\right)}{R_3\left(R_1 + R_2\right)}\left(R_2 v_1 + R_1 v_2\right)$$

Therefore, the output voltage is $\boxed{v_o = \frac{\left(R_3 + R_4\right)}{R_3\left(R_1 + R_2\right)}\left(R_2 v_1 + R_1 v_2\right)}$.